

Experiment 'p_3_2_classify_real' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_3_2_classify_real

Natural language statement: null

Method signature: p_3_2_classify_real (x: real) returns (s: string)

Ensures

- $x == 0.0 \implies s == \text{"zero, small"}$
- $1.0 > x > 0.0 \implies s == \text{"positive, small"}$
- $1.0 \leq x < 1000000.0 \implies s == \text{"positive"}$
- $1000000.0 \leq x \implies s == \text{"positive, large"}$
- $0.0 > x > -1.0 \implies s == \text{"negative, small"}$
- $-1.0 \geq x > -1000000.0 \implies s == \text{"negative"}$
- $-1000000.0 \geq x \implies s == \text{"negative, large"}$

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_3_2_classify_real (x: real) returns (s: string)`

The method should respect the following contract:

```
ensures x == 0.0 ==> s == "zero, small", ensures 1.0 > x > 0.0 ==> s == "positive, small  
  ↳ ", ensures 1.0 <= x < 1000000.0 ==> s == "positive", ensures 1000000.0 <= x ==> s  
  ↳ == "positive, large", ensures 0.0 > x > -1.0 ==> s == "negative, small", ensures  
  ↳ -1.0 >= x > -1000000.0 ==> s == "negative", ensures -1000000.0 >= x ==> s == "  
  ↳ negative, large"
```

Produce and show only the Dafny body of this method, including the curly braces that
 ↳ surround it. Do not show the signature nor contract. You must not use 'assume' nor
 ↳ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn34qglfPvbEfiHyOxJa7OVs2vhMG

Model: gpt-5.5-2026-04-23

Created at: 1780582272

Finish reason: stop

Usage: 494 tokens in, and 603 tokens out

```
{  
  if x == 0.0 {  
    s := "zero, small";  
  } else if x > 0.0 {  
    if x < 1.0 {  
      s := "positive, small";  
    } else if x < 1000000.0 {  
      s := "positive";  
    } else {  
      s := "positive, large";  
    }  
  } else {  
    if x > -1.0 {  
      s := "negative, small";  
    } else if x > -1000000.0 {  
      s := "negative";  
    } else {
```

```
    s := "negative, large";
  }
}
}
```

Final Program

```
method p_3_2_classify_real (x: real) returns (s: string)
  ensures x == 0.0 ==> s == "zero, small"
  ensures 1.0 > x > 0.0 ==> s == "positive, small"
  ensures 1.0 <= x < 1000000.0 ==> s == "positive"
  ensures 1000000.0 <= x ==> s == "positive, large"
  ensures 0.0 > x > -1.0 ==> s == "negative, small"
  ensures -1.0 >= x > -1000000.0 ==> s == "negative"
  ensures -1000000.0 >= x ==> s == "negative, large"
{
  if x == 0.0 {
    s := "zero, small";
  } else if x > 0.0 {
    if x < 1.0 {
      s := "positive, small";
    } else if x < 1000000.0 {
      s := "positive";
    } else {
      s := "positive, large";
    }
  } else {
    if x > -1.0 {
      s := "negative, small";
    } else if x > -1000000.0 {
      s := "negative";
    } else {
      s := "negative, large";
    }
  }
}
```

Total Token Usage

Input tokens: 494

Output tokens: 603

Reasoning tokens: 423

Sum of 'total tokens': 1097

Experiment Timings

Overall Experiment started at 1780582270650, ended at 1780582320074, lasting 49424ms (49.42 seconds)

Iteration #1 started at 1780582270660, ended at 1780582320071, lasting 49411ms (49.41 seconds)